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Compositions and methods for regulating production of an antibody like protein and ribonucleic acid

Abstract

The present disclosure relates to one or more agents, therapies, treatments, and methods of use of the agents and/or therapies and/or treatments for producing an antibody like protein (ALP) and one or more sequences of miRNA that are complementary to the mRNA of a target, viral-specific protein or proteins. Embodiments of the present disclosure can be used as a therapy or a treatment for a subject that has a condition whereby the production of the APL and decreased production of a target, viral-specific protein or proteins may be of therapeutic benefit.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The application is a continuation filing and claims the benefit of U.S. Provisional Patent Application Ser. No. 18/446,965, filed Aug. 8, 2023, the entire contents thereof are incorporated herein by reference.

SEQUENCE LISTING
(1) This application contains a Sequence Listing electronically submitted via Patent Center to the United States Patent and Trademark Office as an XML Document file entitled “A8148706US—ST26.xml” created on 2023 Aug. 8 and having a size of 14,542 bytes. The information contained in the Sequence Listing is incorporated by reference herein.

TECHNICAL FIELD

(2) The present disclosure generally relates to compositions and methods for regulating production of an antibody-like protein (ALP) and ribonucleic acid (RNA). In particular, the present disclosure relates to compositions and methods for regulating gene expression and, therefore, production of an ALP and interfering RNA (iRNA), which may suppress viral infections.

BACKGROUND

(3) Viral infections cause mortality in patients. In particular, viral infections affect patients with suppressed immune systems, for example resulting from illness or aging.

(4) It may be desirable to improve therapies and treatments for patients with viral infections.

SUMMARY

(5) Some embodiments of the present disclosure relate to compositions and methods that upregulate the production of both an antibody-like protein (ALP) that targets a surface protein of a virus and one or more sequences of micro-interfering RNA (miRNA) that is complimentary to and degrades, or causes degradation of, mRNA of a target, viral-specific protein or proteins.

(6) In some embodiments of the present disclosure, the target virus is an influenza A virus. In some embodiments of the present disclosure, the target viral-specific protein is an influenza A virus protein or proteins. Without being bound by any particular theory, the ALP can recognize and bind to one or more surface proteins of the target virus, for example influenza A.

(7) In some embodiments of the present disclosure, the composition comprises a plasmid of deoxyribonucleic acid (DNA) that includes an insert sequence of nucleic acids that encode for the production of the ALP, one or more an insert sequences of nucleic acids that encode for the production of miRNA and a backbone sequence of nucleic acids that facilitate introduction of the insert sequence into one or more of a subject's cells where the insert sequence is expressed and/or replicated. Expression of the insert sequence by one or more cells of the subject results in production of the ALP and production of the miRNA. The production of the ALP within one or more of the subject's cells can then bioactivate, recognize and bind to a surface protein of the infecting virus. The production of miRNA within one or more of the subject's cell may result in decreased translation of a target, viral-specific protein by one or more of the subject's cells.

(8) In some embodiments of the present disclosure, the methods that upregulate the production of the ALP and the one or more miRNA sequences also relate to methods of manufacturing and administering the composition.

(9) Some embodiments of the present disclosure relate to a pharmaceutical agent that comprises an agent, and/or a pharmaceutically acceptable carrier. Administering the pharmaceutical agent to a subject may increase the subject's production of the ALP, and one or more sequences of miRNA that decreases the production of a viral specific protein or proteins.

(10) Some embodiments of the present disclosure relate to compositions and methods that can be used as a therapy or a treatment for a viral infection in a subject.

(11) Some embodiments of the present disclosure relate to a method of treating a viral infection. The method comprises a step of administering to a subject a therapeutically effective amount of an agent that upregulates the subject's production of the ALP, and one or more sequences of miRNA that decreases the production of a target, viral-specific protein or proteins.

(12) Embodiments of the present disclosure relate to at least one method for inducing endogenous production of the ALP and one or more sequences of miRNA that target the mRNA of a viral-specific protein or proteins. One such method utilizes gene vectors containing nucleotide sequences for the production of the ALP and one or more sequences of miRNA that target the mRNA of a viral specific protein or proteins, which can be administered to a subject to produce of the ALP and one or more sequences of miRNA. Without being bound by any particular theory, embodiments of the present disclosure may be useful for treating viral infections.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features of the present disclosure will become more apparent in the following detailed description in which reference is made to the appended drawings.

(2) FIG. **1** is a Kaplan Meier graph of % of animals surviving after infection with influenza A versus time obtained from mice treated with either a control or one of three adeno associated virus (AAV) expression cassettes: one AAV expression cassette with SEQ ID NO: 1; one AAV expression cassette with SEQ ID NO: 2; and, one AAV expression cassette with SEQ ID NO: 3 inserted, according to embodiments of the present disclosure.

DETAILED DESCRIPTION

(3) Unless defined otherwise, all technical and scientific terms used herein have the meanings that would be commonly understood by one of skill in the art in the context of the present description. Although any methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present disclosure, the preferred methods and materials are now described. All publications mentioned herein are incorporated herein by reference to disclose and describe the methods and/or materials in connection with which the publications are cited.

(4) As used herein, the singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. For example, reference to “an agent” includes one or more agents and reference to “a subject” or “the subject” includes one or more subjects.

(5) As used herein, the terms “about” or “approximately” refer to within about 25%, preferably within about 20%, preferably within about 15%, preferably within about 10%, preferably within about 5% of a given value or range. It is understood that such a variation is always included in any given value provided herein, whether or not it is specifically referred to.

(6) As used herein, the term “agent” refers to a substance that, when administered to a subject, causes one or more chemical reactions and/or one or more physical reactions and/or or one or more physiological reactions and/or one or more immunological reactions in the subject. In some embodiments of the present disclosure, the agent is a plasmid, a viral vector containing a plasmid, a protein coat containing a plasmid, or a lipid vesicle containing a plasmid.

(7) As used herein, the term “antibody like protein” refers to proteins that have the same binding properties as “monoclonal antibodies” and may be referred to as “antibody like protein” or “monoclonal antibody” interchangeably.

(8) As used herein, the term “cell” refers to a single cell as well as a plurality of cells or a population of the same cell type or different cell types. Administering an agent to a cell includes in vivo, in vitro and ex vivo administrations and/or combinations thereof.

(9) As used herein, the term “complex” refers to an association, either direct or indirect, between one or more particles of an agent and one or more target cells. This association results in a change in the metabolism of the target cell. As used herein, the phrase “change in metabolism” refers to an increase or a decrease in the one or more target cells' production of DNA, RNA, one or more proteins, and/or any post-translational modifications of one or more proteins.

(10) As used herein, the term “endogenous” refers to the production and/or modification of a molecule that originates within a subject.

(11) As used herein, the terms “inhibit”, “inhibiting”, and “inhibition” refer to a decrease in activity, response, or other biological parameter of a biologic process, disease, disorder or symptom thereof. This can include but is not limited to the complete ablation of the activity, response, condition, or disease. This may also include, for example, a 10% reduction in the activity, response, condition, or disease as compared to the native or control level. Thus, the reduction can be a 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, or any amount of reduction in between the specifically recited percentages, as compared to native or control levels.

(12) As used herein, the term “patient” refers to a subject that is afflicted with a disease or disorder. The term “patient” includes human and veterinary subjects.
(13) As used herein, the term “pharmaceutical composition” means any composition comprising, but not necessarily limited to, an agent to be administered a subject in need of therapy or treatment of a disease, disorder or symptom thereof. Pharmaceutical compositions may also additionally include one or more further active ingredients such as antimicrobial agents, anti-inflammatory agents, anaesthetics, analgesics, and the like.
(14) As used herein, the phrases “prevention of” and “preventing” refer to avoiding the onset or progression of a disease, disorder, or a symptom thereof.
(15) As used herein, the terms “production”, “producing” and “produce” refer to the synthesis and/or replication of DNA, the transcription of one or more sequences of RNA, the translation of one or more amino acid sequences, the post-translational modifications of an amino acid sequence, and/or the production of one or more regulatory molecules that can influence the production and/or functionality of an effector molecule or an effector cell. For clarity, “production” is also be used herein to refer to the functionality of a regulatory molecule, unless the context reasonably indicates otherwise.
(16) As used herein, the terms “promote”, “promotion”, and “promoting” refer to an increase in an activity, response, condition, disease process, or other biological parameter. This can include, but is not limited to, the initiation of the activity, response, condition, or disease process. This may also include, for example, a 10% increase in the activity, response, condition, or disease as compared to the native or control level. Thus, the increase in an activity, response, condition, disease, or other biological parameter can be 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, or more, including any amount of increase in between the specifically recited percentages, as compared to native or control levels.
(17) As used herein, the term “subject” refers to any therapeutic target that receives the agent. The subject can be a vertebrate, for example, a mammal including a human. The term “subject” does not denote a particular age or sex. The term “subject” also refers to one or more cells of an organism, an in vitro culture of one or more tissue types, an in vitro culture of one or more cell types, ex vivo preparations, and/or a sample of biological materials such as tissue and/or biological fluids.
(18) As used herein, the term “target cell” refers to one or more cells and/or cell types that are deleteriously affected, either directly or indirectly, by a disease process. The term “target cell” also refers to cells that are not deleteriously affected but that are the cells in which it is desired that the agent interacts.
(19) As used herein, the term “therapeutically effective amount” refers to the amount of the agent used that is of sufficient quantity to ameliorate, treat and/or inhibit one or more of a disease, disorder or a symptom thereof. The “therapeutically effective amount” will vary depending on the agent used, the route of administration of the agent and the severity of the disease, disorder or symptom thereof. The subject's age, weight and genetic make-up may also influence the amount of the agent that will be a therapeutically effective amount.
(20) As used herein, the terms “treat”, “treatment” and “treating” refer to obtaining a desired pharmacologic and/or physiologic effect. The effect may be prophylactic in terms of completely or partially preventing an occurrence of a disease, disorder or symptom thereof and/or the effect may be therapeutic in providing a partial or complete amelioration or inhibition of a disease, disorder, or symptom thereof. Additionally, the term “treatment” refers to any treatment of a disease, disorder, or symptom thereof in a subject and includes: (a) preventing the disease from occurring in a subject which may be predisposed to the disease but has not yet been diagnosed as having it; (b) inhibiting the disease, i.e., arresting its development; and (c) ameliorating the disease.
(21) As used herein, the terms “unit dosage form” and “unit dose” refer to a physically discrete unit that is suitable as a unitary dose for patients. Each unit contains a predetermined quantity of the

agent and optionally, one or more suitable pharmaceutically acceptable carriers, one or more excipients, one or more additional active ingredients, or combinations thereof. The amount of agent within each unit is a therapeutically effective amount.

(22) In embodiments of the present disclosure, the pharmaceutical compositions disclosed herein comprise an agent as described above in a total amount by weight of the composition of about 0.1% to about 95%. For example, the amount of the agent by weight of the pharmaceutical composition may be about 0.1%, about 0.2%, about 0.3%, about 0.4%, about 0.5%, about 0.6%, about 0.7%, about 0.8%, about 0.9%, about 1%, about 1.1%, about 1.2%, about 1.3%, about 1.4%, about 1.5%, about 1.6%, about 1.7%, about 1.8%, about 1.9%, about 2%, about 2.1%, about 2.2%, about 2.3%, about 2.4%, about 2.5%, about 2.6%, about 2.7%, about 2.8%, about 2.9%, about 3%, about 3.1%, about 3.2%, about 3.3%, about 3.4%, about 3.5%, about 3.6%, about 3.7%, about 3.8%, about 3.9%, about 4%, about 4.1%, about 4.2%, about 4.3%, about 4.4%, about 4.5%, about 4.6%, about 4.7%, about 4.8%. about 4.9%, about 5%, about 5.1%, about 5.2%, about 5.3%, about 5.4%, about 5.5%, about 5.6%, about 5.7%, about 5.8%, about 5.9%, about 6%, about 6.1%, about 6.2%, about 6.3%, about 6.4%, about 6.5%, about 6.6%, about 6.7%, about 6.8%, about 6.9%, about 7%, about 7.1%, about 7.2%, about 7.3%, about 7.4%, about 7.5%, about 7.6%, about 7.7%, about 7.8%, about 7.9%, about 8%, about 8.1%, about 8.2%, about 8.3%, about 8.4%, about 8.5%, about 8.6%, about 8.7%, about 8.8%, about 8.9%, about 9%, about 9.1%, about 9.2%, about 9.3%, about 9.4%, about 9.5%, about 9.6%, about 9.7%, about 9.8%, about 9.9%, about 10%, about 11%, about 12%, about 13%, about 14%, about 15%, about 16%, about 17%, about 18%, about 19%, about 20%, about 25%, about 30%, about 35%, about 40%, about 45%, about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90% or about 95%.

(23) Where a range of values is provided herein, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the disclosure. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges, and are also, encompassed within the disclosure, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the disclosure.

(24) In some embodiments of the present disclosure, an agent is a plasmid for introducing genes into a target cell for reproduction or transcription of an insert carried within the plasmid. In some embodiments of the current disclosure, the plasmid is contained in a lipid vesicle, a protein coat, or combinations of both lipid and protein. In some embodiments of the present disclosure, the plasmid is contained within a viral vector. In some embodiments of the present disclosure, the viral vector is an adeno-associated virus (AAV) vector.

(25) In some embodiments of the present disclosure, the insert comprises one or more nucleotide sequences that encode for production of at least an antibody-like protein (ALP) that targets a surface protein of a virus and one or more sequences of micro-interfering RNA (miRNA) each sequence complementary to the mRNA of a target, viral-specific protein or proteins.

(26) In some embodiments of the present disclosure, the ALP targets a surface protein of a virus, such as a coat protein, a spike protein, a membrane fusion protein or a combination thereof of the by recognizing a portion or all of the primary structure (amino acid sequence), the secondary structure (localized protein structures) or the tertiary structure (the three-dimensional shape) of the surface protein. As such, if one or more of the primary, secondary or tertiary structure of a surface protein is known and if that surface protein is accessible by the ALP, then the ALP can target and bind to the given viral protein. Without being bound by any particular theory, the ALP may act in a similar fashion to an antibody. For example, in some embodiments of the present disclosure the ALP may act like a neutralizing antibody and in other embodiments of the present disclosure the ALP may act like a non-neutralizing antibody.

(27) In some embodiments of the present disclosure, the one or more sequences of miRNA are complementary to and, therefore, bind to the target mRNA and cause the target mRNA to be degraded. The degrading of the target mRNA decreases translation of the target mRNA into a resultant target, viral-specific protein.
(28) The present disclosure relates to one or more agents, therapies, treatments, and methods of use of the agents and/or therapies and/or treatments for initiating or upregulating production of the ALP while downregulating production and/or functionality of the target viral protein or proteins. Some embodiments of the present disclosure relate to methods for making a complex between at least one particle of an agent and at least one target cell of a subject for initiating production of the ALP and for downregulating the subject's production and/or functionality of the target, viral-specific protein or proteins. Embodiments of the present disclosure can be used as a therapy or a treatment for a subject that has a viral infection.
(29) In some embodiments of the present disclosure, the agent can be administered to the subject by an intravenous route, an intramuscular route, an intraperitoneal route, an intrathecal route, an intravesical route, a topical route, an intranasal route, a transmucosal route, a pulmonary route, and combinations thereof.
(30) Some embodiments of the present disclosure relate to an agent that can be administered to a subject with a viral infection. When a therapeutically effective amount of the agent is administered to the subject, the subject may produce an ALP and one or more miRNAs that bind to and cause degradation of the mRNA of one or more target, viral-specific proteins.
(31) In some embodiments of the present disclosure, administering a therapeutic amount of the agent to a subject upregulates the production, functionality or both of the ALP and one or more sequences of miRNA that each bind to and cause degradation of the mRNA of one or more target, viral-specific proteins. In some embodiments of the present disclosure, there are one, two or three miRNA sequences that each are complimentary to and degrade, or cause degradation of the mRNA that can be translated into a viral specific protein or proteins.
(32) In some embodiments of the present disclosure, the agent is a vector used for gene therapy. The gene therapy is useful for inducing the subject's endogenous production of the ALP and one or more sequences of miRNA that target the mRNA of a target, viral-specific protein or proteins. For example, the vector can contain one or more nucleotide sequences that that cause production of the ALP and production of one or more miRNA sequences to inhibit a viral infection.
(33) In some embodiments of the present disclosure, the vector used for gene therapy is a virus that can be a double stranded (ds) DNA virus, a single stranded (ss) DNA virus, a ssRNA virus, a dsRNA virus, or combinations thereof.
(34) The embodiments of the present disclosure also relate to administering a therapeutically effective amount of the agent. In some embodiments of the present disclosure, the therapeutically effective amount of the agent that is administered to a patient is between about 10 and about 1×10.sup.16 TCID.sub.50/kg (50% tissue culture infective dose per kilogram of the patient's body weight). In some embodiments of the present disclosure, the therapeutically effective amount of the agent that is administered to the patient is about 1×10.sup.13 TCID.sub.50/kg. In some embodiments of the present disclosure, the therapeutically effective amount of the agent that is administered to a patient is measured in TPC/kg (total particle count of the agent per kilogram of the patient's body weight). In some embodiments the therapeutically effective amount of the agent is between about 10 and about 1×10.sup.16 TCP/kg.
(35) Some embodiments of the present disclosure relate to a therapy, or method of treating a viral infection, that can be administered to a subject with the viral infection. The therapy comprises a step of administering to the subject a therapeutically effective amount of an agent that will upregulate the subject's production of the ALP and one or more sequences of miRNA that target the mRNA of a viral-specific protein or proteins. The production of the ALP and production of the miRNA may reduce deleterious effects of the viral infection upon the subject.

(36) The embodiments of the present disclosure relate to assisting subject's experiencing a viral infection. While the examples below relate to influenza A, the following categories include member viruses that are also contemplated as target viruses that are treatable with the embodiments of the present disclosure: a double stranded (ds) DNA virus, a single stranded (ss) DNA virus, a ssRNA virus, a dsRNA virus, or combinations thereof.
(37) Below are examples of nucleotide sequences of each may be present in the insert:
(38) SEQ ID NO: 1 (nucleotide sequence that is codon optimized for ALP-Flu20)
(39) TABLE-US-00001
GGTACCGCCACCATGGCTACTGGGTCAAGAACATCTCTGCTGCTGGCTT
TCGGGCTGCTGTGCCTGCCTTGGCTGCAGGAGGGGAGTGCTCAGGTCCA
GCTGCAGGAGAGCGGACCAGGCCTGGTGAAGCCTTCCGAGACACTGTCT
CTGACCTGCTCCGTGTCTGGCGTGTCCGTGACATCTGACATCTACTATT
GGACCTGGATCAGGCAGCCACCTGGCAAGGGCCTGGAGTGGATCGGCTA
CATCTTCTATAACGGCGACACCAACTACAATCCCAGCCTGAAGTCCAGA
GTGACAATGAGCATCGATACCTCCAAGAATGAGTTCTCTCTGAGGCTGA
CAAGCGTGACCGCAGCAGACACAGCCGTGTACTTTTGCGCCAGGGGCAC
CGAGGATCTGGGCTATTGCAGCTCCGGCTCCTGTCCTAACCACTGGGGC
CAGGGCACACTGGTGACCGTGTCTAGCTCCACAAAGGGCCCAAGCGTGT
TTCCTCTGGCCCCATCTAGCAAGAGCACATCCGGAGGCACCGCCGCCCT
GGGATGTCTGGTGAAGGATTACTTCCCAGAGCCCGTGACCGTGTCTTGG
AACAGCGGCGCCCTGACATCCGGAGTGCACACCTTTCCAGCCGTGCTGC
AGTCCTCTGGCCTGTACAGCCTGAGCTCCGTGGTGACAGTGCCTTCTAG
CTCCCTGGGCACACAGACCTATATCTGCAACGTGAATCACAAGCCCAGC
AATACCAAGGTGGACAAGAAGGTGGAGCCTAAGTCCTGTGATAAGACAC
ACACCTGCCCACCATGTCCTGCACCAGAGCTGCTGGGAGGACCATCCGT
GTTCCTGTTTCCTCCAAAGCCCAAGGACACACTGATGATCTCTCGCACA
CCCGAGGTGACCTGCGTGGTGGTGGACGTGAGCCACGAGGATCCTGAGG
TGAAGTTCAACTGGTACGTGGATGGCGTGGAGGTGCACAATGCCAAGAC
CAAGCCTAGAGAGGAGCAGTACAACAGCACATATCGGGTGGTGTCCGTG
CTGACCGTGCTGCACCAGGACTGGCTGAACGGCAAGGAGTATAAGTGCA
AGGTGTCCAATAAGGCCCTGCCCGCCCCTATCGAGAAGACAATCTCTAA
GGCAAAGGGACAGCCAAGGGAGCCTCAGGTGTACACCCTGCCCCCTTCC
AGGGAGGAGATGACAAAGAACCAGGTGTCTCTGACCTGTCTGGTGAAGG
GCTTCTATCCTTCCGACATCGCCGTGGAGTGGGAGTCTAATGGCCAGCC
AGAGAACAATTACAAGACCACACCACCCGTGCTGGACTCCGATGGCTCT
TTCTTTCTGTATTCTAAGCTGACCGTGGATAAGAGCAGATGGCAGCAGG
GCAACGTGTTTTCTTGTAGCGTGATGCACGAGGCCCTGCACAATCACTA
CACACAGAAGTCCCTGTCTCTGAGCCCAGGCAAGAGGAAGAGGAGAGCA
CGAGGCCCTGCACAATCACTACACACAGAAGTCCCTGTCTCTGAGCCCA
GGCAAGAGGAAGAGGAGATCCGGATCTGGAGCACCAGTGAAGCAGACCC
TGAACTTCGACCTGCTGAAGCTGGCCGGCGATGTGGAGAGCAATCCAGG
CCCCATGGCCACAGGCAGCAGAACCTCCCTGCTGCTGGCCTTTGGCCTG
CTGTGCCTGCCATGGCTGCAGGAGGGAAGCGCCGACATCCAGATGACCC
AGTCCCCATCTAGCCTGAGCGCCTCCATCGGCGATCGGGTGACAATCAC
CTGTCGCCCCTCCCAGAACATCAGGTCTTTCCTGAATTGGTTTCAGCAC
AAGCCAGGCAAGGCCCCCAAGCTGCTGATCTACGCAGCATCTAACCTGC
AGAGCGGCGTGCCATCCCGCTTCTCTGGAAGCGGATCCGGCACAGAGTT
TACACTGACCATCAGGTCCCTGCAGCCCGAGGACTTCGCCACCTACTAT
TGCCAGCAGAGCTATAACACACCTCCAACCTTTGGCCAGGGCACAAAGG
TGGAGATCAAGGGACAGCCTAAGGCAGCACCATCCGTGACCCTGTTCCC

ACCTTCCTCTGAGGAGCTGCAGGCCAATAAGGCCACCCTGGTGTGCCTG
ATCAGCGACTTTTACCCTGGAGCAGTGACCGTGGCATGGAAGGCCGATA
GCTCCCCTGTGAGGCCGGCGTGGAGACAACAACCCCATCTAAGCAGAGC
AACAATAAGTACGCCGCCTCTAGCTATCTGTCTCTGACCCCAGAGCAGT
GGAAGAGCCACCGGTCTTATAGCTGTCAGGTGACCCATGAAGGCTCAAC
TGTGGAGAAAACCGTCGCCCCAACTGAATGTTCCTAA

(40) SEQ ID NO: 2 (nucleotide sequence for a codon optimized for miRNA against RNA polymerase, nucleoprotein, and hemagglutinin, each of which is a target, viral-specific protein)

(41) TABLE-US-00002

CTGGAGGCTTGCTGAAGGCTGTATGCTGAAGGATCTTATTTCTTCGGAG
AGTTTTGGCCTCTGACTGACTTTCCGAAGATAAGATCCTTCAGGACACA
AGGCCTGTTACTAGCACTCACATGGAACAAATGGCCTCTAGCCTGGAGG
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(42) SEQ ID NO: 3 (AAV expression cassette with SEQ ID 1 and SEQ ID 2 inserted)

(43) TABLE-US-00003

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Example 1—Expression Cassette

(44) Expression cassettes for expression of a monoclonal antibody (mAb) and/or miRNA were synthesized by Genscript. Each cassette contained a signal peptide, and/or the variable heavy domain, the human IgG1 constant domain, and/or the miRNA sequence followed by (when it is an Ab), a self-cleaving 2A peptide sequence, a signal peptide, the variable light domain and the human lambda constant domain. The synthesized mAb and/or miRNA expression cassettes were cloned

into the pAVA-00200 plasmid backbone containing the CASI promoter1, multiple cloning site (MCS), Woodchuck Hepatitis Virus post-transcriptional regulatory element (WPRE), Simian virus 40 (SV40) polyadenylation (polyA) sequence all flanked by the AAV2 inverted terminal repeats (ITR). pAVA-00200 was cut with the restriction enzymes KpnI and XbaI in the MCS and separated on a 1% agarose gel. The band of interest was excised and purified using a gel extraction kit. Each mAb and/or protein and/or miRNA expression cassette was amplified by PCR using Taq polymerase and the PCR products were gel purified and the bands on interest were also excised and purified using a gel extraction kit. These PCR products contained the mAb and/or protein and/or miRNA expression cassettes in addition to 15 base pair 5′ and 3′ overhangs that align with the ends of the linearized pAVA-00200 backbone. Using in-fusion cloning2, the amplified mAb or protein or miRNA expression cassettes are integrated with the pAVA-00200 backbone via homologous recombination. The resulting plasmid vectors contained the following: 5′ ITR, CASI promoter, and/or mAb expression cassette, and/or miRNA expression cassette, WPRE, SV40 polyA and ITR 3′.

Example 2—Animal Studies

(45) Female BALB/c mice were purchased from Charles River. AAV vectors that included a nucleotide sequence causing expression of an ALP that encodes for production of ALP-Flu20 (Vector 1 that includes SEQ ID No: 1), three sequences of miRNA that bind to and cause degradation of target, viral-specific proteins miRNA's (Vector 2 that includes SEQ ID NO: 2 and that targets the mRNA of each of RNA polymerase, nucleoprotein, and hemagglutinin), or both of the ALP and the three miRNAs (Vector 3 that includes SEQ ID NO: 3), as in Example 1, were administered to 6-week-old BALB/c mice. All animal experiments were approved by the institutional animal care committees of the Canadian Science Centre for Human and Animal Health and the University of Guelph. Intranasal administration of the AAV vectors were performed using a 40-μL injection volume. The dose used was about 2×10.sup.11 vector genomes per mouse. After four weeks the mice were infected intra-nasally with influenza A strain PR8. The viral dose used was 1×10.sup.5 plaque forming units (PFUs).

Example 3—Experimental Data

(46) FIG. **1** presents the data, in the form of a Kaplan-Meir graph, obtained from mice that received an intranasal administration of a sham carrier (control, shown as **10** in FIG. **1**) or one of Vector 1 (shown as **12** in FIG. **1**, Vector 2 (shown as **14** in FIG. **1**) or Vector 3 (shown as **16** in FIG. **1**) followed by intranasal administration influenza A strain PR8.

(47) As shown in FIG. **1**, the group of mice that received Vector 3 (that included sequences that encoded for increased production of ALP and miRNA against the mRNA of three target, viral-specific proteins) demonstrated statistically significant higher survival (p=0.007) than the group of control mice, the group of mice that received Vector 1 and the group of mice that received Vector 2.

(48) Without being bound by any particular theory, the embodiments of the present disclosure that cause a subject to increase production of both the ALP and the one or more miRNA sequences can be used to treat a subject who is infected with a target virus. It is understood that the description and examples provided herein refer to one given target virus, namely influenza A, yet the embodiments of the present disclosure are not limited to just the one exemplary, target virus. For example, a member of the Orthomyxoviridae family of viruses can be a target protein but members of other virus families can also be the target virus. Because the ALP can be designed to target and/or recognize any known surface-protein of any virus and because each of the one or more sequences of miRNA can be designed to be complementary to any known target, viral-specific protein, the embodiments of the present disclosure can be designed for any target virus with at least one known surface protein and at least one target, virus-specific protein with a known mRNA nucleotide sequence.

Claims

1. A method of treating a subject with a condition, the method comprising a step of administering to the subject a therapeutic dose of a composition that comprises a recombinant plasmid (RP) with a sequence of nucleotides that comprise a start region, an end region and an insert positioned between the start region and the end region, wherein the insert encodes for production of an antibody-like protein (ALP) that is bindable with a surface protein of an influenza virus and a further sequence of micro-interfering ribonucleic acid (miRNA) that binds to and causes degradation of messenger ribonucleic acid (mRNA) that encodes for a target, viral protein of the influenza virus, wherein the sequence of nucleotides of the insert comprises: i) a sequence that comprises SEQ ID No: 1, and ii) the further sequence that comprises SEQ ID No: 2.
2. The method of claim 1, wherein the condition is an infection of the influenza virus.